MIL-I-81803(AS)
14 December 1970
Superseding AS-1075
20 September 1968

MILITARY SPECIFICATION

IGNITION ELEMENT, ELECTRICAL, MARK 14 MOD 0

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This document governs the manufacture, assembly and preparation for delivery of the Ignition Element, Electrical, MARK 14 MOD 0, and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this document to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-I-23659	Initiators, Electric, Design and Evaluation of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 1377

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MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

DL 2520046	Ignition Element, Electrical, MARK 14 MOD 0
2519878	Composition, Primer 878 (Formerly Code Ident 10001)
2520046	Ignition Element, Electrical, MARK 14 MOD 0, Assembly
2520053	Electrode Assembly
2520058	Ignition Composition

PUBLICATIONS

Bureau of Naval Weapons (Code Ident 10001)

OP 5	Ammunition Ashore (This publication is not furnished to bidders and contractors by the procuring activity. Contractors should contact the Government Inspector for use of the publication.)
WR-43	Preparation of Quality Assurance Provisions

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

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CODE OF FEDERAL REGULATIONS

49 CFR 171-178	Transportation, Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should request the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 General.

3.1.1 The Ignition Element, Electrical, MARK 14 MOD 0, covered by this specification shall be manufactured in accordance with the drawings listed on DL 2520046.

3.2 Materials. All materials used in the manufacture of the ignition elements shall conform strictly with the specifications referred to on the respective Naval Air Systems Command drawings, unless specific approval in writing covering a departure therefrom has been obtained from the Navy cognizant field activity for cartridges prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Preproduction sample. When specified by the procuring activity, a preproduction sample of 112 ignition elements shall be manufactured using the same methods and procedures proposed for the production lot. The sample shall be tested as specified in 4.3.2 and is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract.

3.3 Primary components. For the purposes of this specification the ignition composition, primer composition, and booster charge are defined as primary components. See 4.3.3.

3.4 Cleanliness and condition of parts. All parts shall be free of chips, dirt, grease, rust and all other foreign materials. There shall be no burrs nor other imperfections of manufacture which would harmfully affect either the assembly or the functioning of the product. The cleaning methods which are used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents which are used.

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3.5 Explosive components, volatiles. The ignition and primer compositions shall have a maximum moisture content of 0.5% at the time of loading, as determined in accordance with 4.7.1.

3.6 Insulation resistance. Prior to assembly of the bridgewire, the insulation resistance of the Electrode Assembly, Drawing 2520053, shall be a minimum of 20 megohms when measured in accordance with 4.7.3 at a relative humidity of 80% or less.

3.7 Bridgewire. The bridgewire shall be spot welded as specified on Drawing 2520053.

3.8 Igniter compositions, compounding and loading. The ignition composition and primer composition shall be compounded in accordance with Drawings 2520058 and 2519878, respectively, and loaded in accordance with Drawing 2520046.

3.9 Booster and ignition charges.

3.9.1 Quantity. The booster charge and consolidated ignition charges shall be as specified on Drawing 2520046. Charges shall be determined by weight rather than by volume measurements.

3.9.2 Smokeless powder, handling and loading. Do not subject the smokeless powder to unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.9.2.1 Smokeless powder, condition. The smokeless powder utilized shall exhibit no evidence of decomposition. See 4.7.2.

3.10 Ignition element inspection.

3.10.1 Visual inspection. Ignition elements shall be free of the following visible defects: chips, burrs, rough edges, improper knurl, improper crimp, perforated closure disc, or any other imperfections which could harmfully affect either assembly or functioning. See 4.7.6.

3.10.2 Bridge circuit resistance. The resistance of the bridge circuit in each completely assembled ignition element, when measured electrically as specified in 4.7.5, shall be not less than 1.0 ohm nor more than 1.2 ohms.

3.10.3 Leakage. Ignition element leakage rate shall not exceed 10^{-5} cc of air or gas per second, when tested in accordance with 4.7.7.

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3.10.4 X-ray. Ignition elements shall be free of the following defects when examined by X-ray in accordance with 4.7.4: improper assembly of components, charges or inert components missing, or any other imperfections which could adversely affect performance.

3.11 Tests. The requirements of 3.11 apply only when government procurement is for the ignition element as an entity. Test requirements for the ignition element when procured installed in cartridges are incorporated in the applicable cartridge specification.

3.11.1 Electric sensitivity.

3.11.1.1 No-fire. The ignition element shall not fire when an electric current of $1.0\ ^{+0.10}_{-0.00}$ ampere is applied in accordance with 4.7.8.1.

3.11.2 Ballistic requirements. Ignition elements from each production lot, selected in accordance with 4.7.9, shall meet the requirements listed below when subjected to the tests prescribed in 4.7.9.

3.11.2.1 Chamber pressure. The Lower (L) limit of maximum pressure shall be: L = 1800 psig.

3.11.2.2 Response time. The time from application of firing current to start of pressure rise shall not exceed 0.015 second.

3.11.2.3 Misfire. There shall be no misfires. See 4.7.9.2.3.

3.11.3 Change in propellant lot. Whenever a change occurs in the propellant lot, test firings shall be conducted in accordance with 4.10. No ignition element shall produce results outside the limits of 3.11.2.1 through 3.11.2.3.

3.12 Packaging, packing and marking. The packaging and packing of ignition elements and the container markings shall be in accordance with Section 5. See 4.6.1.

3.13 Workmanship. Ignition elements shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes and coatings.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-414 or MIL-STD-105, whichever applies.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with MIL-STD-414 or MIL-STD-105, whichever applies.

4.3.2 Preproduction sample. When specified by the procuring activity and before entering into quantity production, an acceptable preproduction sample of 112 ignition elements shall be prepared. Of these ignition elements, 102 shall be expended in the tests listed below and ten shall be retained for investigative purposes. A preproduction sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of paragraphs 3.3 through 3.10 of this specification. Preproduction sample acceptance shall consist of the no-fire test of 4.7.8 and the following tests as prescribed in Table 3 of MIL-I-23659: 40-foot drop (6), temperature and humidity cycling (36), vibration (12), +225°F ± 5°F (12), -80°F ± 5°F (12) and +70°F ± 5°F (12). The number in parentheses denotes the number of ignition elements allocated for each test. In addition, 12 ignition elements shall be subjected to an electrostatic sensitivity test. Each of these 12 ignition elements shall be tested with the stimulus from an electrically charged capacitor, connected between the electrode and case of the ignition element, as specified in the electrical characteristics requirements of MIL-I-23659. No ignition element shall fire when subjected to the electrostatic sensitivity test nor when subjected to the no-fire test of 4.7.8.

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Ignition element firings shall be conducted using the test procedures of 4.7.9 and no ignition element shall produce results outside the limits set forth in 3.11.2.1 through 3.11.2.3. The failure of any ignition element to meet the acceptance requirements as stated above shall cause rejection of the preproduction sample.

4.3.3 Production lot. The suggested ignition element production lot size is 5,000 minimum and 10,000 maximum. Only primary components from a single lot or batch shall be used in a production lot of ignition elements; however, one primary component production lot may be used in more than one ignition element production lot.

4.4 Gages. The contractor shall provide gages as necessary to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

4.5 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.6 Quality conformance inspection. Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of DL 2520046 and verification of the following requirements and tests:

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Item	Nature of Requirement Verification	Applicable Paragraphs	Classification (WR-43)
Primary Component	Single lot or batch	3.3, 4.3.3	(M101)
Ignition and Primer Compositions	Moisture content	3.5, 4.7.1	(C1)
Electrode Assembly	Insulation resistance	3.6, 4.7.3	(M102)
Electrode Assembly	Bridgewire welding	3.7	(C2)
Ignition and Primer Compositions	Compounding and loading	3.8	(C3)
Booster and Ignition Charges	Type and weight	3.9.1	(C4)
Smokeless Powder	Condition	3.9.2.1, 4.7.2	(C5)
Ignition Element	Visual inspection	3.10.1, 4.7.6	(M103)
Ignition Element	Bridge circuit resistance	3.10.2, 4.7.5	(C6)
Ignition Element	Leakage	3.10.3, 4.7.7	(C7)
Ignition Element	X-ray	3.10.4, 4.7.4	(C8)
Ignition Element	Electric sensitivity	3.11.1, 4.7.8	(C9)
Ignition Element	Ballistic requirements	3.11.2, 4.7.9	(C10)
Ignition Element	Ballistics (change in propellant lot)	3.11.3, 4.10	(C11)
Ignition Element	Packaging, packing and container markings	3.12, 4.6.1	(M104)

4.6.1 Packing, packaging and marking. Quality conformance inspection shall be conducted to verify that the packaging, packing and the container markings are in accordance with Section 5.

4.7 Tests.

4.7.1 Determing moisture content. Transfer approximately two grams of the ignition composition to be used in loading to a tared weighing dish, weigh to the nearest 0.1 milligram, calculate the weight of the sample, transfer weighing dish and sample to a vacuum drying oven, dry at +160 ± 5°F and 28 ± 2 inches of mercury vacuum until constant weight is obtained. The loss of weight represents the percentage of moisture in the original sample. Use the same procedure for the primer composition.

4.7.2 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable paragraphs of OP 5. Evidence of decomposition shall be cause for rejection.

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4.7.3 Insulation resistance. Prior to assembly of the bridgewire in the Electrode Assembly, Drawing 2520053, the insulation resistance between the metal parts of the assembly shall be checked with an applied voltage of 500 ± 25 volts DC. Units which fail to meet the requirements of 3.6 shall be rejected.

4.7.4 X-ray of ignition element. Examine all ignition elements by X-ray and reject those having any of the defects of 3.10.4. Ignition elements shall be mounted in suitable racks or frames and laid on their side so that the most revealing exposure can be obtained. Pressure-sensitive tape may be used as an aid in positioning the ignition elements.

4.7.5 Bridge circuit resistance. Conformance with the bridge circuit resistance of 3.10.2 shall be ascertained by means of a test circuit which limits the bridge current to 25 milliamperes, maximum. The test circuit shall be connected to the ignition element through suitable test leads. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable bridge which is known to be accurate within one percent in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirements of 3.10.2 shall be rejected. The results of bridge circuit resistance measurements shall not be affected by resistance of the test circuit and leads.

4.7.6 Visual inspection. Examine all ignition elements visually and reject those having any of the defects of 3.10.1.

4.7.7 Leakage. Each ignition element in the preproduction sample or production sample shall be leak tested in a dry gas or dry air leak tester. Ignition elements which exhibit a leak rate in excess of 10^{-5} cc/sec gas or air at a pressure differential of 1.0 ± 0.1 atmosphere shall be considered defective. Ignition elements of the preproduction sample which exhibit a leakage defect shall be rejected. If one or more ignition elements of the production sample exhibit a leakage defect, then a 100 percent leak test of the lot being tested shall be required at no additional expense to the government. Ignition elements which exhibit a leak rate in excess of that specified above shall be rejected.

4.7.8 Electric sensitivity tests.

4.7.8.1 No-fire test. Each ignition element of the preproduction sample and the production lot sample shall be conditioned at +70 ± 5°F for a minimum of six hours and then tested. During the test insure that the ignition element is sufficiently shielded to prevent personnel injury or material damage in the event an ignition element fires. The firing

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circuit shall contain a suitable resistor in series to regulate the current to 1.0 $^{+0.10}_{-0.00}$ amperes measured to an accuracy of one percent. This current shall be applied to the ignition element for a period of not less than 5 minutes nor more than 5.1 minutes. An ignition element which fires shall be considered defective and shall cause rejection of the lot represented.

4.7.9 Ballistic tests.

4.7.9.1 The number of ignition elements selected from each production lot for the ballistic tests shall be in accordance with the requirements of Tables A-2 and B-1 of MIL-STD-414, Inspection Level IV and an AQL of 0.10. After all ignition elements of the sample under test have met the no-fire requirements of 4.7.8, the elements shall be conditioned for a minimum of six hours at +70 ± 5°F and fired in a closed chamber having an internal volume, exclusive of ignition element volume, of ten cubic centimeters and adapted for measurement of time from application of current to start of pressure rise and maximum pressure. The ratio of length to diameter of the closed chamber shall be approximately 1:1. The firing current shall be 4.0 $^{+0.10}_{-0.00}$ amperes. In addition to the sample of ignition elements for the above test, ten additional ignition elements shall be furnished from the production lot for investigative purposes.

4.7.9.2 Criteria of acceptability.

4.7.9.2.1 Chamber pressure. The lower limit of maximum pressure shall be as specified in 3.11.2.1. The lot shall be acceptable if the quantity $(\bar{X}-L)/S$ is equal to or greater than the appropriate value of K, at an AQL of 0.10, when computed in accordance with Section B, Part I of MIL-STD-414.

4.7.9.2.2 Response time. The failure of any ignition element to meet the response time requirement of 3.11.2.2 shall cause rejection of the lot represented.

4.7.9.2.3 Misfire. The failure of any ignition element to fire shall result in rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.8 Retest. There shall be no retests.

4.9 Test failure. If test failure is attributable to an assignable cause, excluding the test ignition element, original test results should be discarded and that part of the test reconducted.

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4.10 Change in propellant lot. Whenever a change in propellant lot occurs, the charge shall be verified by firing 36 ignition elements which have been temperature conditioned for a minimum of six hours (12 at -80 ± 5°F, 12 at +70 ± 5°F and 12 at +225 ± 5°F). Results of these firings shall meet the requirements of 3.11.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. Not applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C. When directed by the procuring activity, elements intended for expenditure in preproduction sample testing and for production lot acceptance tests or shipment to point of assembly into cartridges, shall be packaged in accordance with Method III of MIL-P-116 and with the Code of Federal Regulations 49 CFR 171-178. For definition of Level C packaging, see 6.4.

5.2 Packing.

5.2.1 Level A. Not applicable.

5.2.2 Level B. Not applicable.

5.2.3 Level C. When directed by the procuring acitivity, elements intended for expenditure in preproduction sample testing and for production lot acceptance tests or shipment to point of assembly into cartridges, and are packaged in accordance with 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping container shall be in accordance with the Code of Federal Regulations 49 CFR 171-178. For definition of Level C packing, see 6.4.

5.3 Markings.

5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3.2 Normal markings. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

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6. NOTES

6.1 Intended use. This ignition element was originally designed to be used in MARK 124 and MARK 125 impulse cartridges.

6.2 Ordering information. Procurement documents should specify the following:

a. title, number and date of this specification;

b. whether preproduction inspection is required, and if so, specify the test activity;

c. production lot size and test activity (see 4.3.3);

d. items of data required for each preproduction and production lot (see 6.3).

6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order.

6.3.1 Test report. Results of ballistic tests shall be forwarded to the Naval Air Systems Command (AIR-53222) and the procuring activity.

6.3.2 Data cards. The contract or order should state the format, quantity and destination of data cards.

6.4 Definitions. For the purposes of this specification, the following definitions apply.

6.4.1 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to supplier's commercial practice when such meets the requirements of this level.

6.4.2 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

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6.5 Explosives safety precautions. Safety precautions for explosive-loaded items are contained in MIL-STD-1314.

6.5.1 Applicability. The loading, assembly and handling of the explosives, explosive-loaded subassemblies and the finished items covered by this specification involve hazardous operations and therefore require suitable explosives safety precautions. At plants of the Naval Air Systems Command, Department of the Navy, and other Government plants where the Government is responsible for safety, the precautions listed in MIL-STD-1314 are mandatory. These same minimum provisions are recommended for observance by a contractor or manufacturer that explosively loads or assembles the item covered in this specification in order to fulfill his own safety responsibility.

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